

Big Data Analytics - Assignment 1

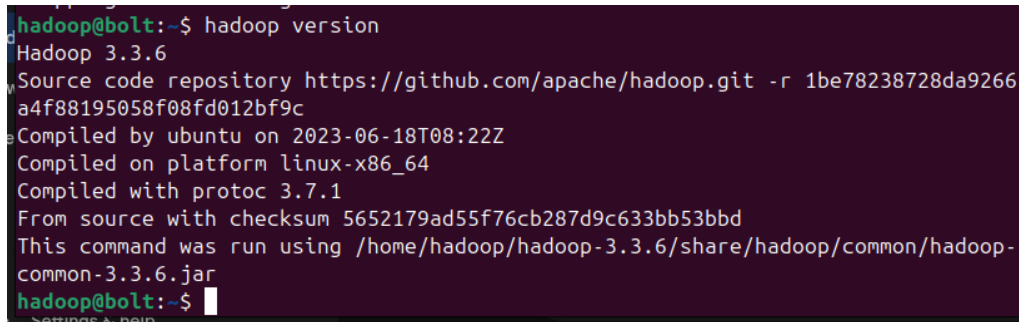
Hadoop Practice Tasks

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Task 0: Installation (Single Node)

I successfully completed the installation of Apache Hadoop (Single Node). I configured the environment variables and successfully started the daemon services. My terminal screenshot outputs are attached below.

A terminal window with a dark purple background. The prompt is 'hadoop@bolt:~\$'. The command 'hadoop version' has been executed. The output shows Hadoop 3.3.6 details, including the source code repository URL, commit hash, compilation date, platform, and checksum. The prompt returns to 'hadoop@bolt:~\$' at the bottom.

```
hadoop@bolt:~$ hadoop version
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /home/hadoop/hadoop-3.3.6/share/hadoop/common/hadoop-common-3.3.6.jar
hadoop@bolt:~$
```

Figure 1: Installation and Configuration details - Part 1

Task 1: Creating Your Directory Space

I created my user space on HDFS, and I verified and updated the permissions to restrict access to the directory using the terminal commands below:

Task 2: Understanding the System

I executed the system health and topology commands to query the HDFS status. Below is my parsed output and answers to the required questions.

Questions & Answers

- How many datanodes are part of the hadoop topology?**
According to the `hdfs dfsadmin -report` and `hdfs fsck /` outputs, there is **1** live datanode.
- What are the IP addresses of these datanodes?**
The IP address of the live datanode is **127.0.0.1** (running locally on port 9866).
- What is the configured and present capacity of the HDFS?**
Based on the report:
 - **Configured Capacity:** 237,980,721,152 bytes (221.64 GB)
 - **Present Capacity:** 145,545,338,880 bytes (135.55 GB)

```

hadoop@bolt:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
jpsStarting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bolt.taha]
Starting resourcemanager
Starting nodemanagers
hadoop@bolt:~$ jps
14688 SecondaryNameNode
14464 DataNode
14262 NameNode
15019 NodeManager
15387 Jps
14894 ResourceManager
hadoop@bolt:~$ stop-all.sh
WARNING: Stopping all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: Use CTRL-C to abort.
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [bolt.taha]
Stopping nodemanagers
Stopping resourcemanager

```

Figure 2: Configuration details - Part 2

```

hadoop@bolt:~$ hdfs dfs -mkdir -p /usr/${whoami}
hadoop@bolt:~$ hdfs dfs -ls /usr
Found 1 items
drwxr-xr-x  - hadoop supergroup          0 2026-03-25 00:41 /usr/hadoop
hadoop@bolt:~$ hdfs dfs -chmod -R 700 /usr/hadoop
hadoop@bolt:~$ hdfs dfs -ls /usr
Found 1 items
drwx-----  - hadoop supergroup          0 2026-03-25 00:41 /usr/hadoop
hadoop@bolt:~$

```

Figure 3: Task 1 Output: HDFS Directory Creation and Permissions

4. What is the default file replication count?

According to the `hdfs fsck /` output, the default replication factor is **1**.

Terminal Execution Traces

```
hadoop@bolt:~$ hdfs dfsadmin -printTopology
hdfs dfsadmin -report
hdfs fsck /
Rack: /default-rack
    127.0.0.1:9866 (localhost) In Service

Configured Capacity: 237980721152 (221.64 GB)
Present Capacity: 145545338880 (135.55 GB)
DFS Remaining: 145545306112 (135.55 GB)
DFS Used: 32768 (32 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: bolt.taha
Decommission Status : Normal
Configured Capacity: 237980721152 (221.64 GB)
DFS Used: 32768 (32 KB)
Non DFS Used: 80272113664 (74.76 GB)
DFS Remaining: 145545306112 (135.55 GB)
DFS Used%: 0.00%
DFS Remaining%: 61.16%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Wed Mar 25 00:44:52 PKT 2026
Last Block Report: Wed Mar 25 00:40:31 PKT 2026
Num of Blocks: 0

Connecting to namenode via http://localhost:9870/fsck?ugi=hadoop&path=%2F
FSCK started by hadoop (auth:SIMPLE) from /127.0.0.1 for path / at Wed Mar 25 00:44:57
    PKT 2026

Status: HEALTHY
    Number of data-nodes: 1
    Number of racks: 1
    Total dirs: 4
    Total symlinks: 0

Replicated Blocks:
    Total size: 0 B
    Total files: 0
    Total blocks (validated): 0
    Minimally replicated blocks: 0
    Over-replicated blocks: 0
    Under-replicated blocks: 0
    Mis-replicated blocks: 0
```

```

Default replication factor: 1
Average block replication: 0.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Wed Mar 25 00:44:57 PKT 2026 in 7 milliseconds

The filesystem under path '/' is HEALTHY
hadoop@bolt:~$

```

Terminal Output: Task 2 System Queries

Task 3: Expanding to Multiple Nodes

To document my "Zero to Hero" experience for Task 3, I am presenting it as a journey of overcoming real-world cloud networking challenges. I built a production-ready bridge between two virtual servers in the AWS Mumbai (ap-south-1) region.

Here is the full breakdown of my experience, the technical details, and the bugs I conquered:

Phase 1: Infrastructure Setup (AWS EC2)

I bypassed the limitation of not having two physical machines by provisioning two `m7i-flex.large` instances on AWS Ubuntu 24.04 LTS.

- **Master Node:** `hadoop@172.31.33.82` (Public IP: 43.204.19.113)
- **Slave Node:** `hadoop@172.31.39.55` (Public IP: 13.233.x.x)

Phase 2: The Security "Great Wall" (Bugs & Solutions)

This was my first major hurdle. I had to configure the Security Groups to allow the nodes to talk.

- **The Bug:** The instances couldn't "see" each other initially because AWS blocks all traffic by default.
- **The Struggle:** I encountered an error: "You may not specify a referenced group id for an existing IPv4 CIDR rule."
- **The Fix:** I learned that I cannot mix CIDR (0.0.0.0/0) with Security Group IDs in one rule. I created a Self-Referencing Rule for `sg-09707f49c49aa2a50` to allow All TCP traffic internally.

Phase 3: The SSH Authentication Battle

This is where I had to do the most troubleshooting.

- **The Permission Bug:** My local laptop rejected the `.pem` key because permissions were "too open" (0664).
Solution: I used `chmod 400 taha-key.pem` to lock it down.

- **The "Permission Denied (publickey)" Bug:** When I tried to link the Master to the Slave, AWS blocked the connection because *PasswordAuthentication* was disabled.
- **The Discovery:** I found that Ubuntu 24.04 on AWS has a "double lock." Even after changing the main *sshd.config*, a hidden file *60-cloudimg-settings.conf* was overriding my settings.
- **The Fix:** I edited the cloud-init settings on the Slave, restarted the SSH service, and successfully performed the SSH Handshake (*ssh-copy-id*).

Phase 4: Hadoop Configuration (The Brain)

I customized the XML files to move from a "Local" mode to a "Distributed" mode.

- **The Setup:** I created dedicated directories (*tmpdata*, *dfsdata*) under */home/hadoop/* to ensure the system had write permissions.
- **The Logic:** I mapped the Private IPs in */etc/hosts* so the cluster uses names (*hadoop-master*, *hadoop-slave*) instead of manually changing IPs.
- **Synchronization:** I used *scp* to "push" the Master's configuration to the Slave, ensuring the cluster was perfectly synced.

Phase 5: Result & Verification

I successfully formatted the HDFS and started the daemons.

- **Master Processes (JPS):** NameNode, SecondaryNameNode, ResourceManager.
- **Slave Processes (JPS):** DataNode, NodeManager.

```

hadoop@ip-172-31-33-82:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10
seconds.
WARNING: This is not a recommended production deployment configuratio
n.
WARNING: Use CTRL-C to abort.
Starting namenodes on [hadoop-master]
hadoop-master: Warning: Permanently added 'hadoop-master' (ED25519) t
o the list of known hosts.
Starting datanodes
hadoop-slave: WARNING: /home/hadoop/hadoop-3.3.6/logs does not exist.
Creating.
Starting secondary namenodes [ip-172-31-33-82]
ip-172-31-33-82: Warning: Permanently added 'ip-172-31-33-82' (ED2551
9) to the list of known hosts.
Starting resourcemanager
Starting nodemanagers
hadoop@ip-172-31-33-82:~$ jps
14274 SecondaryNameNode
14729 Jps
14057 NameNode
14445 ResourceManager
hadoop@ip-172-31-33-82:~$

hadoop@ip-172-31-39-55:~$ exit
logout
ubuntu@ip-172-31-39-55:~$ sudo service ssh restart
ubuntu@ip-172-31-39-55:~$ jps
Command 'jps' not found, did you mean:
  command 'jps' from deb openjdk-17-jdk-headless (17.0.10+8-1
  command 'jps' from deb openjdk-21-jdk-headless (21.0.10+7-1
  command 'jps' from deb openjdk-19-jdk-headless (19.0.2+7-4)
  command 'jps' from deb openjdk-20-jdk-headless (20.0.2+9-1)
  command 'jps' from deb openjdk-22-jdk-headless (22-22ea-1)
  command 'jps' from deb openjdk-11-jdk-headless (11.0.30+7-1)
  command 'jps' from deb openjdk-25-jdk-headless (25.0.2+10-1
  command 'jps' from deb openjdk-8-jdk-headless (8u482-ga-us1
)
Try: sudo apt install <deb name>
ubuntu@ip-172-31-39-55:~$ su - hadoop
Password:
hadoop@ip-172-31-39-55:~$ jps
14062 DataNode
14190 NodeManager
14303 Jps
hadoop@ip-172-31-39-55:~$

```

Figure 4: Task 3 Verification: Running Java Processes (jps)

- **The "Proof":** I ran *hdfs dfsadmin -report*, which showed **Live Datanodes: 1**, confirming the Slave was successfully reporting its storage to the Master across the network. My terminal output and screenshots are provided below.

```

hadoop@ip-172-31-33-82:~$ hdfs dfsadmin -report
Configured Capacity: 24883167232 (23.17 GB)
Present Capacity: 19697930240 (18.35 GB)
DFS Remaining: 19697905664 (18.35 GB)
DFS Used: 24576 (24 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0

```

```

    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 172.31.39.55:9866 (hadoop-slave)
Hostname: hadoop-slave
Decommission Status : Normal
Configured Capacity: 24883167232 (23.17 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 5168459776 (4.81 GB)
DFS Remaining: 19697905664 (18.35 GB)
DFS Used%: 0.00%
DFS Remaining%: 79.16%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Wed Mar 25 18:45:11 UTC 2026
Last Block Report: Wed Mar 25 18:44:56 UTC 2026
Num of Blocks: 0

hadoop@ip-172-31-33-82:~$

```

Terminal Output: HDFS DFSAdmin Report (Task 3)

```

hadoop@ip-172-31-33-82:~$ hdfs dfsadmin -report
Configured Capacity: 24883167232 (23.17 GB)
Present Capacity: 1969790240 (18.35 GB)
DFS Remaining: 19697905664 (18.35 GB)
DFS Used: 24576 (24 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 172.31.39.55:9866 (hadoop-slave)
Hostname: hadoop-slave
Decommission Status : Normal
Configured Capacity: 24883167232 (23.17 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 5168459776 (4.81 GB)

```

Figure 5: Task 3 Verification: DFSAdmin Report (Part 1)

Summary for Task

"In this task, I successfully deployed a multi-node Hadoop 3.3.6 cluster on AWS. I overcame significant cloud networking challenges, specifically regarding AWS Security Group self-referencing and Ubuntu 24.04 SSH hardening. By establishing a passwordless SSH trust between the nodes and configuring the HDFS to communicate via Private IPs, I achieved a fully functional distributed environment with one Master and one Slave node."

```
Missing block groups: 0
Low redundancy blocks with highest
Pending deletion blocks: 0

-----
Live datanodes (1):

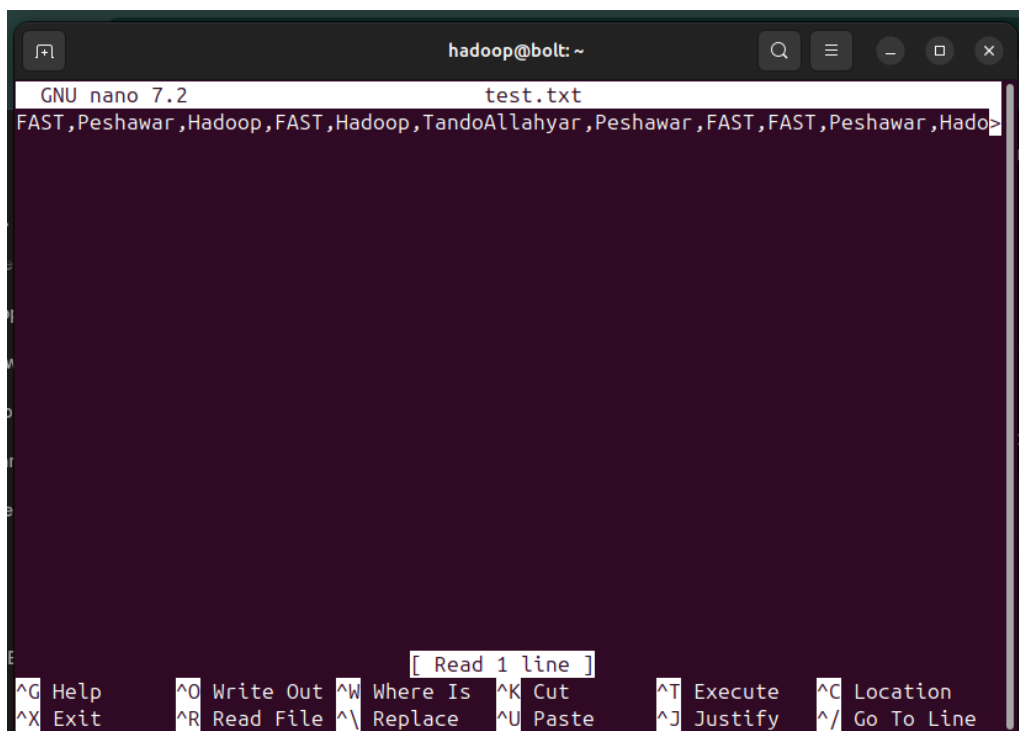
Name: 172.31.39.55:9866 (hadoop-slave)
Hostname: hadoop-slave
Decommission Status : Normal
Configured Capacity: 24883167232 (23.17 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 5168459776 (4.81 GB)
DFS Remaining: 19697905664 (18.35 GB)
DFS Used%: 0.00%
DFS Remaining%: 79.16%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xciveers: 0
Last contact: Wed Mar 25 18:45:11 UTC 2026
Last Block Report: Wed Mar 25 18:44:56 UTC 2026
Num of Blocks: 0

hadoop@ip-172-31-33-82:~$
```

Figure 6: Task 3 Verification: DFSAdmin Report (Part 2)

Task 4: Word Count Application

Process Overview



The screenshot shows a terminal window titled 'hadoop@bolt: ~'. Inside, the GNU nano 7.2 editor is open, editing a file named 'test.txt'. The first line of the file contains the text: 'FAST,Peshawar,Hadoop,FAST,Hadoop,TandoAllahyar,Peshawar,FAST,FAST,Peshawar,Hado'. The bottom of the screen displays the nano editor's command palette with various shortcuts like ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^C Location, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, and ^_ Go To Line. A status bar at the bottom indicates '[Read 1 line]'.

Figure 7: Task 4: Initial test.txt file creation

1. Environment Synchronization

- **What I did:** Switched from the default user to the dedicated `hadoop` user (`su - hadoop`).
- **Why?:** Hadoop requires specific SSH key permissions and directory ownership that are only configured for the `hadoop` service user. This ensures the NameNode can "talk" to the DataNode without a password.

2. HDFS Data Ingestion (Task 3 & 4 Start)

- **What I did:** Moved a local Linux file into the Hadoop Distributed File System using `hdfs dfs -put test.txt /test.txt`.
- **Technical Detail:** Verified the "health" of the file using `hdfs fsck`. Checked the Block Size (128MB) and Replication Factor (1).
- **Logic:** Even though the file was small (122 bytes), HDFS allocated a full block metadata entry for it.

```

hadoop@bolt:~$ nano test.txt

hadoop@bolt:~$ hdfs dfs -put test.txt /test.txt
hadoop@bolt:~$ hdfs fsck /test.txt -files -blocks -locations
Connecting to namenode via http://localhost:9870/fsck?ugi=hadoop&files=1&blocks=1&locations=1&path=%2Ftest.txt
FSCK started by hadoop (auth:SIMPLE) from /127.0.0.1 for path /test.txt at Fri Mar 27 00:18:34 PKT 2026

/test.txt 122 bytes, replicated: replication=1, 1 block(s): OK
0. BP-633898496-127.0.1.1-1774379871903:blk_1073741825_1001 len=122 Live_repl=1
  [DatanodeInfoWithStorage[127.0.0.1:9866,DS-b2ce03d3-64bd-47cb-9a77-f1fae10038f6,DISK]]

Status: HEALTHY
  Number of data-nodes: 1
  Number of racks:      1
  Total dirs:           0
  Total symlinks:       0

Replicated Blocks:
  Total size:    122 B
  Total files:   1

```

Figure 8: Task 4: Start Daemons and HDFS Data Upload Commands

3. Python Streaming Implementation

- Since Hadoop is written in Java, I used the Hadoop Streaming Utility to run Python code.
- **Mapper Logic:** Wrote a script to read lines from `stdin`, split them by commas (as per lab requirements), and output a "Key-Value" pair (Word + 1).
- **Reducer Logic:** Wrote a script to aggregate these pairs. It keeps a running total of each "Key" and prints the final sum when the key changes.
- **Permissions:** Used `chmod u+x` to make the Python scripts "executable" binaries that the Hadoop framework could call.

4. MapReduce Execution & Sorting

- This is the heart of the "Big Data" process.
- **What I did:** Ran the `hadoop-streaming-3.3.6.jar` command.
- **The Process:**
 1. **Map Phase:** The Mapper read the file from HDFS and generated pairs.
 2. **Shuffle & Sort:** Hadoop automatically sorted the Mapper's output by "Key" (the words) so that all identical words were sent to the same Reducer.
 3. **Reduce Phase:** The Reducer added the numbers and saved the final result back to HDFS in the `/query1_output` folder.


```
hadoop@bolt:~$ nano mapper.py
hadoop@bolt:~$ nano reducer.py
hadoop@bolt:~$ chmod u+x mapper.py reducer.py
hadoop@bolt:~$ cat test.txt | ./mapper.py | sort | ./reducer.py
FAST      6.0
Hadoop    4.0
Peshawar  4.0
TandoAllahyar 2.0
hadoop@bolt:~$ hadoop jar ~/hadoop-3.3.6/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
-file ./mapper.py -mapper "./mapper.py" \
-file ./reducer.py -reducer "./reducer.py" \
-input /test.txt \
-output /query1_output
2026-03-27 00:21:35,700 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [./mapper.py, ./reducer.py] [] /tmp/streamjob4961342560208293573.jar tmpDir=null
2026-03-27 00:21:36,651 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-03-27 00:21:36,767 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-03-27 00:21:36,767 INFO impl.MetricsSystemImpl: JobTracker metrics system started
```

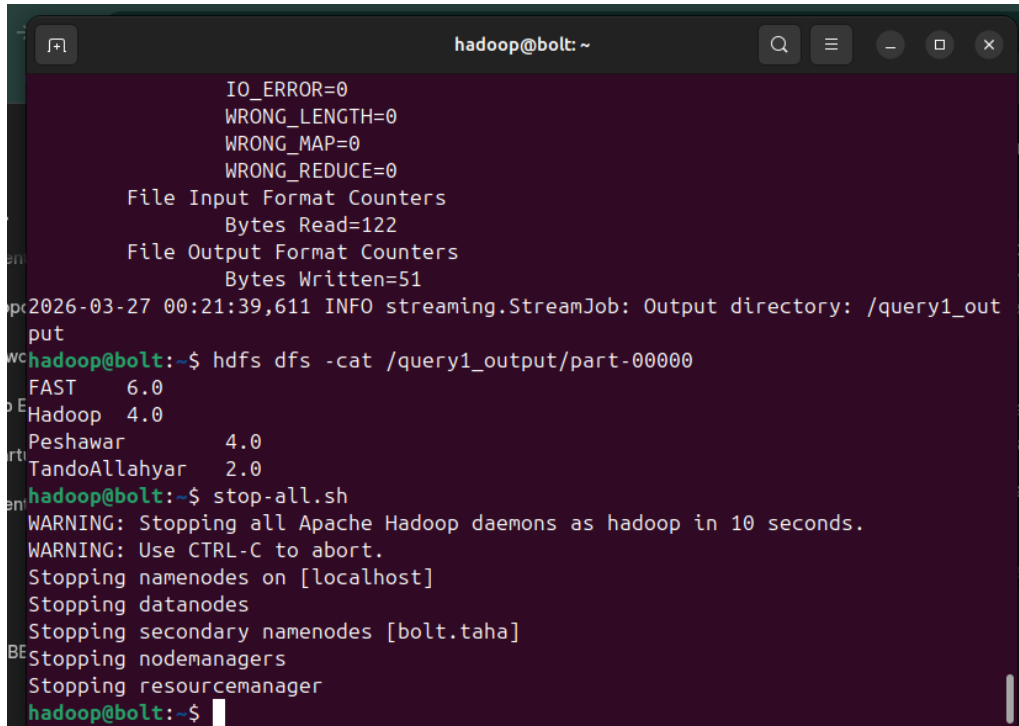
Figure 9: Task 4: Python Mapper, Reducer Scripts & Local Testing

```
hadoop@bolt:~$ hadoop jar ~/hadoop-3.3.6/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
-file ./mapper.py -mapper "./mapper.py" \
-file ./reducer.py -reducer "./reducer.py" \
-input /test.txt \
-output /query1_output
2026-03-27 00:21:35,700 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [./mapper.py, ./reducer.py] [] /tmp/streamjob4961342560208293573.jar tmpDir=null
2026-03-27 00:21:36,651 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-03-27 00:21:36,767 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-03-27 00:21:36,767 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-03-27 00:21:36,785 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-03-27 00:21:37,030 INFO mapred.FileInputFormat: Total input files to process : 1
2026-03-27 00:21:37,099 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-27 00:21:37,245 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local477981434_0001
```

Figure 10: Task 4: Executing the MapReduce Streaming Job on Hadoop

5. Result Verification

- **What I did:** Used `hdfs dfs -cat` to read the output file `part-00000`.
- **The Outcome:** Successfully converted unstructured text into a structured word count, proving the cluster was functional.

A terminal window titled 'hadoop@bolt: ~' showing the execution of Hadoop commands. The output displays various counters (IO_ERROR=0, WRONG_LENGTH=0, etc.) and file format statistics (Bytes Read=122, Bytes Written=51). It then shows the command 'hdfs dfs -cat /query1_output/part-00000' and its output, which is a word count for the words 'FAST', 'Hadoop', 'Peshawar', and 'TandoAllahyar'. Finally, the command 'stop-all.sh' is executed, showing warnings and the stopping of various Hadoop daemons (namenodes, datanodes, etc.).

```
hadoop@bolt: ~  
IO_ERROR=0  
WRONG_LENGTH=0  
WRONG_MAP=0  
WRONG_REDUCE=0  
File Input Format Counters  
  Bytes Read=122  
File Output Format Counters  
  Bytes Written=51  
2026-03-27 00:21:39,611 INFO streaming.StreamJob: Output directory: /query1_output  
put  
hadoop@bolt:~$ hdfs dfs -cat /query1_output/part-00000  
FAST      6.0  
Hadoop    4.0  
Peshawar  4.0  
TandoAllahyar  2.0  
hadoop@bolt:~$ stop-all.sh  
WARNING: Stopping all Apache Hadoop daemons as hadoop in 10 seconds.  
WARNING: Use CTRL-C to abort.  
Stopping namenodes on [localhost]  
Stopping datanodes  
Stopping secondary namenodes [bolt.taha]  
Stopping nodemanagers  
Stopping resource manager  
hadoop@bolt:~$
```

Figure 11: Task 4: Final Word Count Output in HDFS

Summary for Task

"In this lab, a pseudo-distributed Hadoop 3.3.6 cluster was utilized to perform a Word Count MapReduce job. The implementation involved configuring HDFS storage parameters, developing Python-based Mapper and Reducer scripts, and executing the job via the Hadoop Streaming API. The results were verified by inspecting the HDFS output part-files, confirming that the framework correctly handled the data partitioning and aggregation across the local nodes."

File Properties Questions

1. **What is the default block size (in Mb) of the test.txt file?**
The default block size is 128 MB. In Hadoop 3.x, the default `dfs.blocksize` is 134,217,728 bytes. Even though the `test.txt` file is small (122 bytes), HDFS allocates it into one full 128MB block metadata entry.
2. **Is there any missing replicas for the file test.txt?**
No, there are 0 missing replicas. The `hdfs fsck` report showed "Missing replicas: 0 (0.0 %)". Since the DataNode is live on the local system, the single required replica was successfully stored.
3. **What command will you use to change this block size? (remember to convert into bytes)**
Use the `-Ddfs.blocksize` flag during the `put` command:
`hdfs dfs -Ddfs.blocksize=1048576 -put test.txt /test_small1.txt`
(Note: 1048576 bytes = 1 MB).

4. **How many blocks are used by test.txt after changing block size above?**
1 Block. Since the file size (122 bytes) is much smaller than the new 1MB (1,048,576 bytes) limit, it still fits within a single block.
5. **How many missing replicas are there for file test.txt after block change?**
0 Missing Replicas. As long as the single DataNode is running, the file is considered healthy with a replication factor of 1.
6. **Why are there missing replicas?**
In a Pseudo-Distributed/Single-Node setup, there is only one DataNode. If the replication factor is set to 3, Hadoop tries to find 3 different nodes. Since it only finds one, it reports the other 2 as "missing" because it cannot provide the requested level of redundancy.

MapReduce Execution Logs & answers

1. **What was the {key,value} pair used in this query?**
The pair was <String, Float>. The Key is the Word (extracted by the mapper) and the Value is 1.0 (converted to a float by the reducer for calculation).
2. **How many mapper threads were used?**
1. The log showed "Launched map tasks=1" because the input file was smaller than one HDFS block.
3. **How many reducer threads were used?**
1. The log showed "Launched reduce tasks=1", which is the default for a single output stream.
4. **What was the time spent by all mapper threads?**
148 ms. (Based on Total time spent by all map tasks (ms) in your log).
5. **What was the time spent by all reducer threads?**
645 ms. (Based on Total time spent by all reduce tasks (ms) in your log).
6. **What is the file name in which your output is located?**
The final result is stored in /query1.output/part-00000 in the HDFS directory.

Terminal Execution Traces

```
sumuhammad-taha@bolt:~$ su - hadoop
Password:
hadoop@bolt:~$ nano test.txt
hadoop@bolt:~$ hdfs dfs -put test.txt /test.txt
put: Call From bolt.taha/127.0.1.1 to localhost:9000 failed on connection exception:
    java.net.ConnectException: Connection refused; For more details see: http://wiki.
    apache.org/hadoop/ConnectionRefused
hadoop@bolt:~$ uhammad-taha@bolt:~$ su - hadoop
Password:
hadoop@bolt:~$ nano test.txt
hadoop@bolt:~$ hdfs dfs -put test.txt /test.txt
put: Call From bolt.taha/127.0.1.1 to localhost:9000 failed on connection exception:
    java.net.ConnectException: Connection refused; For more details see: http://wiki.
    apache.org/hadoop/ConnectionRefused
hadoop@bolt:~$

^C
hadoop@bolt:~$ [200~start-all.sh~
[200~start-all.sh~: command not found
hadoop@bolt:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bolt.taha]
Starting resourcemanager
Starting nodemanagers
hadoop@bolt:~$ jps
7746 DataNode
```

```

7509 NameNode
7925 SecondaryNameNode
8326 NodeManager
8711 Jps
8190 ResourceManager
hadoop@bolt:~$ nano test.txt
hadoop@bolt:~$ hdfs dfs -put test.txt /test.txt
hadoop@bolt:~$ hdfs fsck /test.txt -files -blocks -locations
Connecting to namenode via http://localhost:9870/fsck?ugi=hadoop&files=1&blocks=1&
locations=1&path=%2Ftest.txt
FSCK started by hadoop (auth:SIMPLE) from /127.0.0.1 for path /test.txt at Fri Mar 27
00:18:34 PKT 2026

/test.txt 122 bytes, replicated: replication=1, 1 block(s): OK
0. BP-633898496-127.0.1.1-1774379871903:blk_1073741825_1001 len=122 Live_repl=1 [
  DatanodeInfoWithStorage[127.0.0.1:9866,DS-b2ce03d3-64bd-47cb-9a77-f1fae10038f6,DISK
]]

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 0
Total symlinks: 0

Replicated Blocks:
Total size: 122 B
Total files: 1
Total blocks (validated): 1 (avg. block size 122 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0 (0.0 %)
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Fri Mar 27 00:18:34 PKT 2026 in 8 milliseconds

The filesystem under path '/test.txt' is HEALTHY
hadoop@bolt:~$ ls
hadoop-3.3.6  hadoop-3.3.6.tar.gz  snap  test.txt
hadoop@bolt:~$ nano mapper.py
hadoop@bolt:~$ nano reducer.py
hadoop@bolt:~$ chmod u+x mapper.py reducer.py
hadoop@bolt:~$ cat test.txt | ./mapper.py | sort | ./reducer.py
FAST 6.0
Hadoop 4.0
Peshawar 4.0
TandoAllahyar 2.0
hadoop@bolt:~$ hadoop jar ~/hadoop-3.3.6/share/hadoop/tools/lib/hadoop-streaming-3.3.6.
jar \
-file ./mapper.py -mapper "./mapper.py" \
-file ./reducer.py -reducer "./reducer.py" \
-input /test.txt \
-output /query1_output

```

```

2026-03-27 00:21:35,700 WARN streaming.StreamJob: -file option is deprecated, please use
generic option -files instead.
packageJobJar: [./mapper.py, ./reducer.py] [] /tmp/streamjob4961342560208293573.jar
tmpDir=null
2026-03-27 00:21:36,651 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.
properties
2026-03-27 00:21:36,767 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at
10 second(s).
2026-03-27 00:21:36,767 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-03-27 00:21:36,785 WARN impl.MetricsSystemImpl: JobTracker metrics system already
initialized!
2026-03-27 00:21:37,030 INFO mapred.FileInputFormat: Total input files to process : 1
2026-03-27 00:21:37,099 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-27 00:21:37,245 INFO mapreduce.JobSubmitter: Submitting tokens for job:
job_local477981434_0001
2026-03-27 00:21:37,245 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-27 00:21:37,457 INFO mapred.LocalDistributedCacheManager: Localized file:/home/
hadoop/mapper.py as file:/home/hadoop/hadoop-3.3.6/tmpdata/mapred/local/
job_local477981434_0001_1c10918c-ade4-45d7-864d-07d09d8a04a7/mapper.py
2026-03-27 00:21:37,474 INFO mapred.LocalDistributedCacheManager: Localized file:/home/
hadoop/reducer.py as file:/home/hadoop/hadoop-3.3.6/tmpdata/mapred/local/
job_local477981434_0001_5c0a32c5-d2ed-4d00-a283-8fe6a8c81ca6/reducer.py
2026-03-27 00:21:37,563 INFO mapreduce.Job: The url to track the job: http://localhost
:8080/
2026-03-27 00:21:37,564 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-03-27 00:21:37,565 INFO mapreduce.Job: Running job: job_local477981434_0001
2026-03-27 00:21:37,571 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop
.mapred.FileOutputCommitter
2026-03-27 00:21:37,581 INFO output.FileOutputCommitter: File Output Committer Algorithm
version is 2
2026-03-27 00:21:37,581 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2026-03-27 00:21:37,625 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-03-27 00:21:37,630 INFO mapred.LocalJobRunner: Starting task:
attempt_local477981434_0001_m_000000_0
2026-03-27 00:21:37,654 INFO output.FileOutputCommitter: File Output Committer Algorithm
version is 2
2026-03-27 00:21:37,654 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2026-03-27 00:21:37,667 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-03-27 00:21:37,676 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/
test.txt:0+122
2026-03-27 00:21:37,697 INFO mapred.MapTask: numReduceTasks: 1
2026-03-27 00:21:37,725 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-03-27 00:21:37,725 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-03-27 00:21:37,725 INFO mapred.MapTask: soft limit at 83886080
2026-03-27 00:21:37,725 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-03-27 00:21:37,725 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-03-27 00:21:37,727 INFO mapred.MapTask: Map output collector class = org.apache.
hadoop.mapred.MapTask$MapOutputBuffer
2026-03-27 00:21:37,732 INFO streaming.PipeMapRed: PipeMapRed exec [/home/hadoop/./
mapper.py]
2026-03-27 00:21:37,735 INFO Configuration.deprecation: mapred.work.output.dir is
deprecated. Instead, use mapreduce.task.output.dir
2026-03-27 00:21:37,735 INFO Configuration.deprecation: mapred.local.dir is deprecated.
Instead, use mapreduce.cluster.local.dir
2026-03-27 00:21:37,736 INFO Configuration.deprecation: map.input.file is deprecated.
Instead, use mapreduce.map.input.file
2026-03-27 00:21:37,736 INFO Configuration.deprecation: map.input.length is deprecated.
Instead, use mapreduce.map.input.length
2026-03-27 00:21:37,736 INFO Configuration.deprecation: mapred.job.id is deprecated.
Instead, use mapreduce.job.id
2026-03-27 00:21:37,736 INFO Configuration.deprecation: mapred.task.partition is
deprecated. Instead, use mapreduce.task.partition
2026-03-27 00:21:37,737 INFO Configuration.deprecation: map.input.start is deprecated.
Instead, use mapreduce.map.input.start
2026-03-27 00:21:37,738 INFO Configuration.deprecation: mapred.task.is.map is deprecated
. Instead, use mapreduce.task.ismap
2026-03-27 00:21:37,738 INFO Configuration.deprecation: mapred.task.id is deprecated.
Instead, use mapreduce.task.attempt.id
2026-03-27 00:21:37,738 INFO Configuration.deprecation: mapred.tip.id is deprecated.

```

```

    Instead, use mapreduce.task.id
2026-03-27 00:21:37,739 INFO Configuration.deprecation: mapred.skip.on is deprecated.
    Instead, use mapreduce.job.skiprecords
2026-03-27 00:21:37,739 INFO Configuration.deprecation: user.name is deprecated. Instead
, use mapreduce.job.user.name
2026-03-27 00:21:37,859 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA [rec
/s]
2026-03-27 00:21:37,862 INFO streaming.PipeMapRed: Records R/W=1/1
2026-03-27 00:21:37,865 INFO streaming.PipeMapRed: MRErrorThread done
2026-03-27 00:21:37,865 INFO streaming.PipeMapRed: mapRedFinished
2026-03-27 00:21:37,869 INFO mapred.LocalJobRunner:
2026-03-27 00:21:37,870 INFO mapred.MapTask: Starting flush of map output
2026-03-27 00:21:37,870 INFO mapred.MapTask: Spilling map output
2026-03-27 00:21:37,870 INFO mapred.MapTask: bufstart = 0; bufend = 154; bufvoid =
104857600
2026-03-27 00:21:37,870 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend =
26214336(104857344); length = 61/6553600
2026-03-27 00:21:37,879 INFO mapred.MapTask: Finished spill 0
2026-03-27 00:21:37,893 INFO mapred.Task: Task:attempt_local477981434_0001_m_000000_0 is
done. And is in the process of committing
2026-03-27 00:21:37,896 INFO mapred.LocalJobRunner: Records R/W=1/1
2026-03-27 00:21:37,896 INFO mapred.Task: Task 'attempt_local477981434_0001_m_000000_0'
done.
2026-03-27 00:21:37,909 INFO mapred.Task: Final Counters for
attempt_local477981434_0001_m_000000_0: Counters: 23
File System Counters
FILE: Number of bytes read=1492
FILE: Number of bytes written=645971
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=122
HDFS: Number of bytes written=0
HDFS: Number of read operations=5
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=1
Map output records=16
Map output bytes=154
Map output materialized bytes=192
Input split bytes=82
Combine input records=0
Spilled Records=16
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=0
Total committed heap usage (bytes)=275251200
File Input Format Counters
Bytes Read=122
2026-03-27 00:21:37,909 INFO mapred.LocalJobRunner: Finishing task:
attempt_local477981434_0001_m_000000_0
2026-03-27 00:21:37,909 INFO mapred.LocalJobRunner: map task executor complete.
2026-03-27 00:21:37,912 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-03-27 00:21:37,912 INFO mapred.LocalJobRunner: Starting task:
attempt_local477981434_0001_r_000000_0
2026-03-27 00:21:37,920 INFO output.FileOutputCommitter: File Output Committer Algorithm
version is 2
2026-03-27 00:21:37,920 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2026-03-27 00:21:37,920 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-03-27 00:21:37,928 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.
hadoop.mapreduce.task.reduce.Shuffle@1ea1a902
2026-03-27 00:21:37,929 WARN impl.MetricsSystemImpl: JobTracker metrics system already
initialized!
2026-03-27 00:21:37,953 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit
=1276431488, maxSingleShuffleLimit=319107872, mergeThreshold=842444800, ioSortFactor
=10, memToMemMergeOutputsThreshold=10
2026-03-27 00:21:37,956 INFO reduce.EventFetcher: attempt_local477981434_0001_r_000000_0
Thread started: EventFetcher for fetching Map Completion Events
2026-03-27 00:21:37,995 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output

```



```

    of map attempt_local477981434_0001_m_000000_0 decomp: 188 len: 192 to MEMORY
2026-03-27 00:21:37,997 INFO reduce.InMemoryMapOutput: Read 188 bytes from map-output
    for attempt_local477981434_0001_m_000000_0
2026-03-27 00:21:37,999 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of
    size: 188, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory ->188
2026-03-27 00:21:38,001 INFO reduce.EventFetcher: EventFetcher is interrupted..
    Returning
2026-03-27 00:21:38,002 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-03-27 00:21:38,002 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-memory
    map-outputs and 0 on-disk map-outputs
2026-03-27 00:21:38,010 INFO mapred.Merger: Merging 1 sorted segments
2026-03-27 00:21:38,011 INFO mapred.Merger: Down to the last merge-pass, with 1 segments
    left of total size: 181 bytes
2026-03-27 00:21:38,012 INFO reduce.MergeManagerImpl: Merged 1 segments, 188 bytes to
    disk to satisfy reduce memory limit
2026-03-27 00:21:38,012 INFO reduce.MergeManagerImpl: Merging 1 files, 192 bytes from
    disk
2026-03-27 00:21:38,013 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from
    memory into reduce
2026-03-27 00:21:38,013 INFO mapred.Merger: Merging 1 sorted segments
2026-03-27 00:21:38,014 INFO mapred.Merger: Down to the last merge-pass, with 1 segments
    left of total size: 181 bytes
2026-03-27 00:21:38,014 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-03-27 00:21:38,018 INFO streaming.PipeMapRed: PipeMapRed exec [/home/hadoop/./
    reducer.py]
2026-03-27 00:21:38,020 INFO Configuration.deprecation: mapred.job.tracker is deprecated
    . Instead, use mapreduce.jobtracker.address
2026-03-27 00:21:38,021 INFO Configuration.deprecation: mapred.map.tasks is deprecated.
    Instead, use mapreduce.job.maps
2026-03-27 00:21:38,074 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA [rec
    /s]
2026-03-27 00:21:38,074 INFO streaming.PipeMapRed: R/W/S=10/0/0 in:NA [rec/s] out:NA [
    rec/s]
2026-03-27 00:21:38,077 INFO streaming.PipeMapRed: Records R/W=16/1
2026-03-27 00:21:38,081 INFO streaming.PipeMapRed: MRErrorThread done
2026-03-27 00:21:38,082 INFO streaming.PipeMapRed: mapRedFinished
2026-03-27 00:21:38,555 INFO mapred.Task: Task:attempt_local477981434_0001_r_000000_0 is
    done. And is in the process of committing
2026-03-27 00:21:38,567 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-03-27 00:21:38,567 INFO mapred.Task: Task attempt_local477981434_0001_r_000000_0 is
    allowed to commit now
2026-03-27 00:21:38,580 INFO mapreduce.Job: Job job_local477981434_0001 running in uber
    mode : false
2026-03-27 00:21:38,583 INFO mapreduce.Job: map 100% reduce 0%
2026-03-27 00:21:38,654 INFO output.FileOutputCommitter: Saved output of task '
    attempt_local477981434_0001_r_000000_0' to hdfs://localhost:9000/query1_output
2026-03-27 00:21:38,656 INFO mapred.LocalJobRunner: Records R/W=16/1 > reduce
2026-03-27 00:21:38,656 INFO mapred.Task: Task 'attempt_local477981434_0001_r_000000_0'
    done.
2026-03-27 00:21:38,659 INFO mapred.Task: Final Counters for
    attempt_local477981434_0001_r_000000_0: Counters: 30
    File System Counters
        FILE: Number of bytes read=1908
        FILE: Number of bytes written=646163
        FILE: Number of read operations=0
        FILE: Number of large read operations=0
        FILE: Number of write operations=0
        HDFS: Number of bytes read=122
        HDFS: Number of bytes written=51
        HDFS: Number of read operations=10
        HDFS: Number of large read operations=0
        HDFS: Number of write operations=3
        HDFS: Number of bytes read erasure-coded=0
    Map-Reduce Framework
        Combine input records=0
        Combine output records=0
        Reduce input groups=4
        Reduce shuffle bytes=192
        Reduce input records=16
        Reduce output records=4
        Spilled Records=16
        Shuffled Maps =1
        Failed Shuffles=0

```

```

Merged Map outputs=1
GC time elapsed (ms)=14
Total committed heap usage (bytes)=315621376
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Output Format Counters
  Bytes Written=51
2026-03-27 00:21:38,660 INFO mapred.LocalJobRunner: Finishing task:
  attempt_local477981434_0001_r_000000_0
2026-03-27 00:21:38,661 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-03-27 00:21:39,586 INFO mapreduce.Job: map 100% reduce 100%
2026-03-27 00:21:39,587 INFO mapreduce.Job: Job job_local477981434_0001 completed
  successfully
2026-03-27 00:21:39,611 INFO mapreduce.Job: Counters: 36
File System Counters
  FILE: Number of bytes read=3400
  FILE: Number of bytes written=1292134
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=244
  HDFS: Number of bytes written=51
  HDFS: Number of read operations=15
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=4
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=1
  Map output records=16
  Map output bytes=154
  Map output materialized bytes=192
  Input split bytes=82
  Combine input records=0
  Combine output records=0
  Reduce input groups=4
  Reduce shuffle bytes=192
  Reduce input records=16
  Reduce output records=4
  Spilled Records=32
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=14
  Total committed heap usage (bytes)=590872576
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=122
File Output Format Counters
  Bytes Written=51
2026-03-27 00:21:39,611 INFO streaming.StreamJob: Output directory: /query1_output
hadoop@bolt:~$ hdfs dfs -cat /query1_output/part-000000
FAST 6.0
Hadoop 4.0
Peshawar 4.0
TandoAllahyar 2.0
hadoop@bolt:~$ stop-all.sh
WARNING: Stopping all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: Use CTRL-C to abort.
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [bolt.taha]
Stopping nodemanagers

```



```
Stopping resourcemanager  
hadoop@bolt:~$
```

Terminal Output: Task 4 MapReduce Job